

STUDY GUIDE FOR MIDTERM II—SOLUTIONS

ECON 101—INTRODUCTION TO MICROECONOMICS

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The following are suggested solutions. Please note that this is only meant for you to check your answers so my solutions are quite succinct; in particular, I do not provide the graphs here.

Not all problems require you to provide a graph. However, my suggestion is that you should always approach any problem you face both graphically and algebraically.

1. (a)
 - i. $P^e = \frac{A-B}{a+b}$
 - ii. $Q^e = \frac{aB+Ab}{a+b}$
 - iii. $P^d = \frac{A+bT-B}{a+b}$
 - iv. $P^s = \frac{-aT+A-B}{a+b}$
 - v. $Q^T = \frac{-abT+aB+Ab}{a+b}$
 - vi. Government revenue is $\frac{T(a(B-bT)+Ab)}{a+b}$
 - vii. Consumer incidence is $\frac{b}{a+b}$ and producer incidence is $\frac{a}{a+b}$
 - viii. $DWL = \frac{abT^2}{2(a+b)}$
- (b)
 - i. $P^d = \frac{A-bS-B}{a+b}$
 - ii. $P^s = \frac{aS+A-B}{a+b}$
 - iii. $Q^S = \frac{abS+aB+Ab}{a+b}$
 - iv. Government expenditure is $\frac{S(a(bS+B)+Ab)}{a+b}$
 - v. Consumer incidence is $\frac{b}{a+b}$ and producer incidence is $\frac{a}{a+b}$, similar to the case of a tax.
 - vi. $DWL = \frac{abS^2}{2(a+b)}$
- (c)
 - i. If $a = 0$ then $P^s = P^e$, which implies that the consumer pays the whole tax/receives the whole subsidy, as can be seen by the consumer incidence expression. Furthermore Q^T and Q^S are equal to Q^e .
 - ii. If $b = 0$ then $P^d = P^e$, which implies that the producer pays the whole amount of the tax/receives the whole subsidy, as can be seen by the producer incidence expression. Furthermore, the dead-weight loss in this case is also zero. Furthermore Q^T and Q^S are equal to Q^e .
 - iii. In both the tax and subsidy cases, the dead-weight loss expression becomes zero if either $a = 0$ or $b = 0$. When demand or supply are perfectly inelastic, the distortion created by the tax or subsidy is zero and therefore the dead-weight loss is zero.
 - iv. As mentioned before, when demand is inelastic, the tax incidence/subsidy incidence expression says that the consumer will pay the whole tax/receive the whole subsidy. Similarly, when supply is inelastic, the tax incidence/subsidy incidence expression says that the producer will pay the whole amount of the tax/receive the whole subsidy.
- (d)
 - i. Consider first the case of perfectly elastic demand. We have that $Q^e = B + bP^*$ and $P^e = P^*$.

	Tax	Subsidy
P^d	P^*	P^*
P^s	$P^* - T$	$P^* + S$
$Q^T Q^S$	$B + bP^* - bT = Q^e - bT$	$B + bP^* + bS = Q^e + bS$
Revenue Expenditure	$T(B + bP^* - bT)$	$S(B + bP^* + bS)$
Consumer Incidence Producer Incidence	0 1	0 1
Dead-Weight Loss	$\frac{bT^2}{2}$	$\frac{bS^2}{2}$

ii. Now consider the case of perfectly elastic supply. We have that $Q^e = A - aP^*$ and $P^e = P^*$.

	Tax	Subsidy
P^d	$P^* + T$	$P^* - S$
P^s	P^*	P^*
$Q^T Q^S$	$A - aP^* + -T = Q^e - aT$	$A - aP^* + aS = Q^e + aS$
Revenue Expenditure	$T(A - aP^* - aT)$	$S(A - aP^* + aS)$
Consumer Incidence Producer Incidence	1 0	1 0
Dead-Weight Loss	$\frac{aT^2}{2}$	$\frac{aS^2}{2}$

2. Table 1 Provides the calculations for taxes while table 2 shows the corresponding calculations for subsidies.

Table 1: Calculations for taxes

	Case I	Case II	Case III	Case IV	Case V
Amount of tax paid by consumers	111.11	1.00	0.00	22,500.00	7.00
Amount of tax paid by producers	222.22	3.00	4,500,000.00	22,500.00	0.00
Government revenue from the tax	333.33	4.00	4,500,000.00	45,000.00	7.00
Consumer surplus before tax	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before tax	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before tax	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after tax	277.78	0.50	81,000,000.00	11,250.00	24.50
Producer surplus after tax	505.56	1.50	4,500,000.00	11,250.00	0.00
Social surplus after tax	783.33	2.00	85,500,000.00	22,500.00	24.50
Dead Weight Loss	33.33	2.00	0.00	22,500.00	0.50

Table 2: Calculations for subsidies

	Case I	Case II	Case III	Case IV	Case V
Amount of subsidy received by consumers	155.56	2.06	0.00	100,000.00	180.00
Amount of subsidy received by producers	311.11	6.19	1,800,000.00	100,000.00	0.00
Government expenditure from the subsidy	466.67	8.25	1,800,000.00	200,000.00	180.00
Consumer surplus before subsidy	400.00	2.00	81,000,000.00	45,000.00	32.00
Producer surplus before subsidy	750.00	6.00	9,000,000.00	45,000.00	0.00
Social surplus before subsidy	1,150.00	8.00	90,000,000.00	90,000.00	32.00
Consumer surplus after subsidy	544.44	3.78	81,000,000.00	125,000.00	162.00
Producer surplus after subsidy	1,038.89	11.34	10,800,000.00	125,000.00	0.00
Social surplus after subsidy	1,583.33	15.13	91,800,000.00	250,000.00	162.00
Dead Weight Loss	33.33	1.13	0.00	40,000.00	50.00

From both tables you can check that the identities indeed hold. Consider the following:

- For a tax, where $SS^{\text{before tax}} \geq SS^{\text{after tax}}$, the change in social surplus is $\Delta SS = SS^{\text{before tax}} - SS^{\text{after tax}}$. Thus, using the identity, we can write

$$\Delta SS - DWL = R,$$

which shows that government revenues are given by the change in social surplus minus the dead-weight loss which shows that indeed some part of the surplus is lost irrevocably.

- For a subsidy, where $SS^{\text{before subsidy}} \leq SS^{\text{after subsidy}}$, the change in social surplus is $\Delta SS = SS^{\text{after subsidy}} - SS^{\text{before subsidy}}$. Thus, using the identity, we can write

$$\Delta SS + DWL = E,$$

which shows that resources from government expenditures are used to fund the change in social surplus but part of these resources are also being lost in the form of an irrevocable loss. v

- $P^d = \frac{(\tau+1)(A-B)}{a(\tau+1)+b}$
 - $P^s = \frac{A-B}{a(\tau+1)+b}$
 - $Q^\tau = \frac{aB(\tau+1)+Ab}{a(\tau+1)+b}$
 - Government revenue is $\frac{\tau(A-B)(aB(\tau+1)+Ab)}{(a\tau+a+b)^2}$
 - $DWL = \frac{ab\tau^2(A-B)^2}{2(a+b)(a\tau+a+b)^2}$
- The aggregate willingness-to-pay curve or *marginal social benefit* of the public good is $MSB = 50 - \frac{1}{2}Q$.
 - You obtain this from the private willingness-to-pay curve: for 50 acres, the maximum each individual is willing to pay is 2.5.
 - Equating MSB to MC yields an optimal size of $Q^* = 50$.
- Notice that individual A is not willing to pay for more than 10 units of the public good. Thus, we need to be careful when vertically adding up the curves. For quantities less than 10, we will add vertically the two marginal benefit schedules. For quantities greater than 10, the marginal social benefit is given only by what individual B is willing to pay.

Therefore, the marginal social benefit curve is

$$MSB = \begin{cases} 22 - \frac{7}{4}Q & 0 \leq Q \leq 10 \\ 12 - \frac{3}{4}Q & 10 \leq Q \leq 16 \end{cases}$$

- Equating MSB with MC yields $Q^* = \frac{22}{3} \approx 7.3$. This solution was obtained by taking the part of the MSB curve between 0 and 10. Notice that the solution we obtained is consistent with the expression for the MSB .
 - Once again, equating MSB with MC yields $Q^* = 12$. This solution was obtained by taking the part of the MSB curve between 10 and 16. The solution that we obtained is consistent with the expression for the MSB . Notice that, if we had erroneously equated MC with the part of the MSB that is between 0 and 10 we would have obtained a solution of 11, which is of course, inconsistent with the actual MSB schedule.
- In this problem, there is an *externality in production* given by the external cost of $MD = 6Q$. Thus, the marginal social cost curve is $MSC = 8Q$. Equating the marginal social benefit curve, which in this case is given by MB , with the MSC curve yields the efficient quantity of $Q^* = 30$. The free market quantity is obtained by equating MC with MB which yields $Q^{\text{market}} = 36$. The market *overproduces* the good. The optimal tax necessary to restore efficiency in the market is obtained by considering the vertical distance between the MSC curve and the MC curve at the efficient quantity. Doing this yields that the tax per unit should be $T = 60$.
 - The marginal benefit curve is $MB = 300 - \frac{Q}{4}$ and the marginal cost curve is $MC = 100 + \frac{Q}{2}$.
 - This good is causing a negative externality. The problem does not specify whether the externality is caused by consuming the good or by producing the good. If the externality is caused by consumption of the good, then we would need to subtract the marginal damage from the MB curve. If the externality is caused by production of the good, then we would need to add the marginal damage to the MC curve. However, since the marginal damage of the externality is constant, from a mathematical point of view, *either way of approaching the problem will yield the same solution*. However, the interpretation would be different. Except in cases where the marginal damage is constant, like in this problem, *you must always be careful whether the externality is caused by consumption or by production of the good*.

For this problem, I am gonna assume that the externality is caused by consuming the good. The marginal social benefit curve is then $MSB = 292 - \frac{Q}{4}$. Equating the previous expression with marginal cost yields an efficient quantity of $Q^* = 256$. The free market quantity is $Q^{\text{market}} = \frac{800}{3} \approx 266.67$. Thus the market overproduces the good by $\frac{32}{3} \approx 10.67$ units.

- (b) At the market quantity, the private marginal benefit is $\frac{700}{3}$ but the marginal social benefit is $\frac{676}{3}$. Thus, the dead-weight loss is given by

$$\frac{(\frac{700}{3} - \frac{676}{3}) \times (\frac{800}{3} - 256)}{2} = \frac{128}{3} \approx 42.67.$$

8. (a) The marginal private benefit is $MB = 1000 - Q$ and the marginal private cost is $MC = Q$. Equating the previous two expressions yields $Q^{\text{market}} = 500$ and $P^{\text{market}} = 500$.
- (b) The marginal social benefit curve is $MSB = 3000 - Q$. Equating this to the marginal cost yields $Q^* = 1500$ and $P^* = 1500$. The efficient quantity could be obtained by different policies. For example giving students grants (like we do next) to subsidize education, or by giving school vouchers worth a monetary amount.
- (c) Let G be the amount of the grant. With the grant the private marginal benefit curve becomes $MB^G = 1000 + G - Q$. The quantity of education demanded after the grant is given to the students is obtained by equating the previous expression with the private marginal cost, which yields

$$Q^G = \frac{1000 + G}{2}.$$

The cost to the government as a function of the grant is then

$$\text{Cost}(G) = 500G + \frac{G^2}{2}.$$

The dead-weight loss is

$$DWL = \frac{|(Q^* - Q^G)(MSB(Q^G) - MB^G(Q^G))|}{2} = \frac{(1000 - G/2)(2000 - G)}{2} = (1000 - G/2)^2.$$

The price paid by families for education is

$$P^d = MB(Q^G) = 500 - \frac{G}{2}.$$

Therefore:

Grant amount	\$1000	\$2000	\$3000
Cost to the government	\$1,000,000	\$3,000,000	\$6,000,000
Dead-weight loss	\$250,000	\$0	\$250,000
After Grant quantity	1000	1500	2000
Achieves efficient outcome?	No (under provision)	Yes	No (over provision)
Price paid my student	0	-500	-1000

With the grant, the consumer pays a price P^d . If this price is negative, this can be interpreted as a situation in which the grant is not only paying in full for the tuition of the student, but it is also paying for the “living costs” of the student.

9. (a) In this case the equilibrium quantity is $Q = 75$, and the price that Jones charges his customers is 10. Consumer surplus is $(15 \times 75)/2 = 562.5$ and Jones surplus, which is the area below the price he charges and above marginal cost, is 0.
- (b) In this case, the equilibrium quantity is $Q = 100$, the price that Jones charges his customers is 10. Consumer surplus is $(20 \times 100)/2 = 1000$. For the dead-weight loss, we need to obtain the marginal social cost curve, which is $MSC = 10 + 0.2Q$. Equating this with the marginal benefit curve when the windows are open yields a quantity of $Q^* = 50$. The marginal social cost at the free market quantity is 30. Thus, the Dead-weight loss is $(50 \times 20)/2 = 500$.

- (c) We calculated this in the last part. The socially efficient quantity when Smith leaves his windows open is $Q^* = 50$.
- (d) This is obtained by looking at the difference between marginal social cost and marginal cost at the efficient quantity. $MSC(50) = 20$ and $MC(50) = 10$. Thus, Jones would compensate an amount of $T = 10$ per customer. Jones would increase his price by 10, i.e. he will charge a price of \$20 which of course yields an equilibrium quantity of 50. Jones total surplus is obtained by looking at his profits, which is the price he charges minus the constant marginal cost times the equilibrium quantity, and then subtracting the total amount he needs to compensate to Smith, which is the T times the equilibrium quantity. Smith's profits are then $(20 - 10) \times 50 = 500$. The total amount of compensation is $10 \times 50 = 500$. Thus Jones' surplus is 0.
- (e) If Jones compensates Smith for opening the windows as described before, then he is indifferent between this scheme and for Smith to leave his windows closed, since in both cases Jones gets a surplus of zero.
10. (a) Equating the private marginal benefit curve to private marginal cost yields $Q^{\text{market}} = 10,000$.
- (b) Equating the marginal social benefit to marginal cost yields $Q^* = 20,000$. Compared to (a), there was an under-provision of flu shots.
- (c) The government could give a subsidy for flu shots. As usual, the required subsidy is given by the vertical distance between the marginal social benefit and the marginal private benefit at the optimal quantity. We have that $MSB(20,000) = 8$ and $MB(20,000) = 3$. Thus, the optimal subsidy that will yield the efficient quantity is to subsidize \$3 per flu shot.
11. (a) $MR = 100 - 2Q$
- (b) Equating MC with MR yields $Q^M = 25$ and a monopoly price $P^M = 75$.
- (c) Revenues are 1875, total costs are 641 so profits are 1234.
- (d) Since marginal cost has not changed, $Q^M = 25$ and $P^M = 75$. However total costs are now 657, so profits are 1218.
- (e) Equating MC with MR yields $Q^M = 10$ and $P^M = 90$. Revenues are 900, total costs are 416 and thus profits are 484.
- (f) Equating MC with MR yields $Q^M = 40$ and $P^M = 60$. The efficient quantity and price are $Q^* = 80$ and $P^* = 20$. The dead-weight loss is 800.
12. Iraq's marginal revenue is $MR_{\text{Iraq}} = 400 - Q$ and Iran's marginal revenue is $MR_{\text{Iran}} = 300 - 2Q$. Segmenting the market and equating the marginal cost to the marginal revenue of each country yields

$$\begin{aligned} Q_{\text{Iraq}}^M &= 200 \\ Q_{\text{Iran}}^M &= 100 \end{aligned}$$

$$\begin{aligned} P_{\text{Iraq}}^M &= 300 \\ P_{\text{Iran}}^M &= 200 \end{aligned}$$

13. (a) Intel faces the market demand and average revenue curve given by $P = 48 - 4X_D$. Thus, its marginal revenue curve is $MR = 48 - 8X_D$. Equating MR with MC yields $X^M = 4$ and the price Intel will charge is $P^M = 32$ per chip.
- (b) In this case we should equate the market demand with marginal cost, so $X^* = 6$ and the efficient price would be $P^* = 24$.
- (c) Since the efficient price is 24, then a price ceiling (price cap) of 24 would make the monopolist have a marginal revenue curve equal to $MR = 24$. In this case, then the monopolist would indeed produce the efficient quantity of 6.
- (d) If we apply a subsidy of s per unit, then the demand curve becomes $P = 48 - 4X_D + s$. Thus, marginal revenue would become $MR = 48 - 8X_D + s$. Note that this makes intuitive sense, since now Intel will get a subsidy of s per unit, then each unit brings an extra revenue of s in addition to what it is already bringing. Intel then equates marginal revenue to marginal cost to get the after subsidy quantity:

$$X^s = \frac{48 + s}{12}.$$

Since the efficient quantity is 6, we can use this to solve for the optimal subsidy to get $s^* = 24$.

- (e) Note that since the monopolist decides how much to produce at its preferred price and when doing so, it always produces in the elastic part of the demand, it is *as if the monopolist was more “inelastic”*. Thus, when giving a subsidy, it goes in full to the monopolist. Therefore price regulation and subsidizing a monopolist are two policies that achieve the same efficient quantity but have very different distributional consequences. The subsidy increases Intel’s profits at the expense of the consumers; the price regulation redistributes some of Intel’s surplus to consumers.